Online POMDP Methods

Approximate POMDP Solutions

Numerical Approximations

(approximately solve original problem)

Offline

Previously

Online

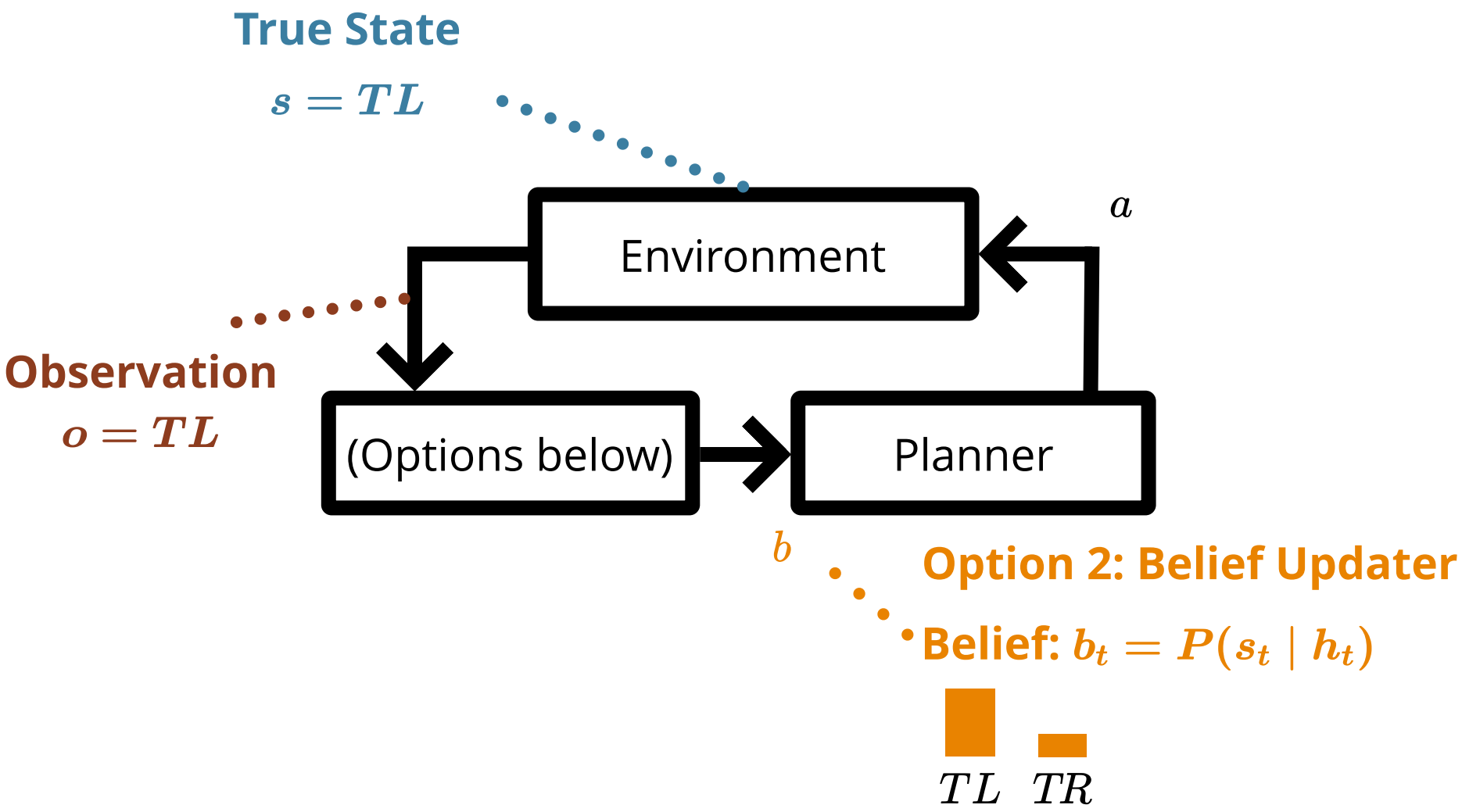
Today!

Formulation Approximations

(solve a slightly different problem)

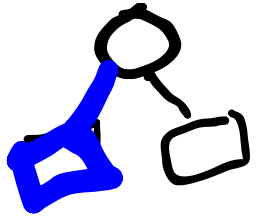
Last Time

POMDP Sense-Plan-Act Loop

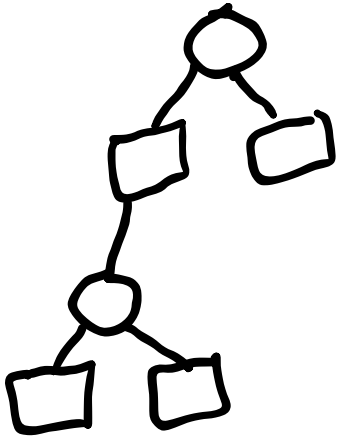


Monte Carlo Tree Search (MCTS/UCT)

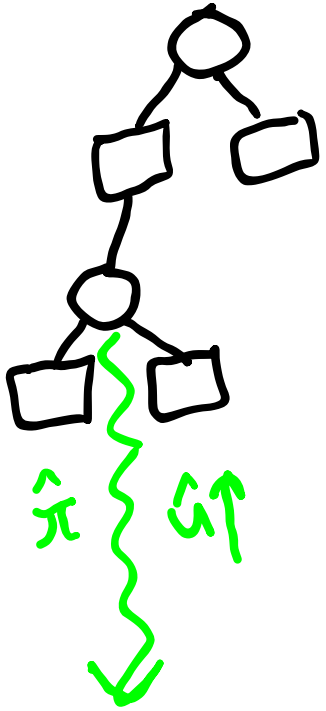
Search



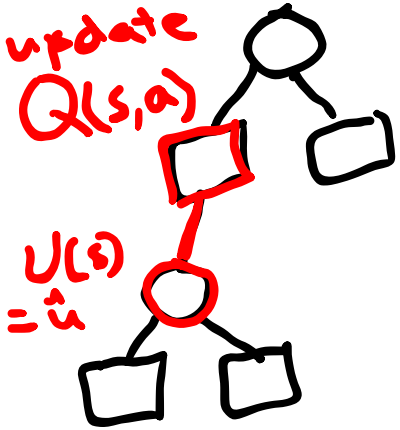
Expansion



Rollout



Backup



$$Q(s,a) + c\sqrt{\frac{\log N(s)}{N(s,a)}}$$

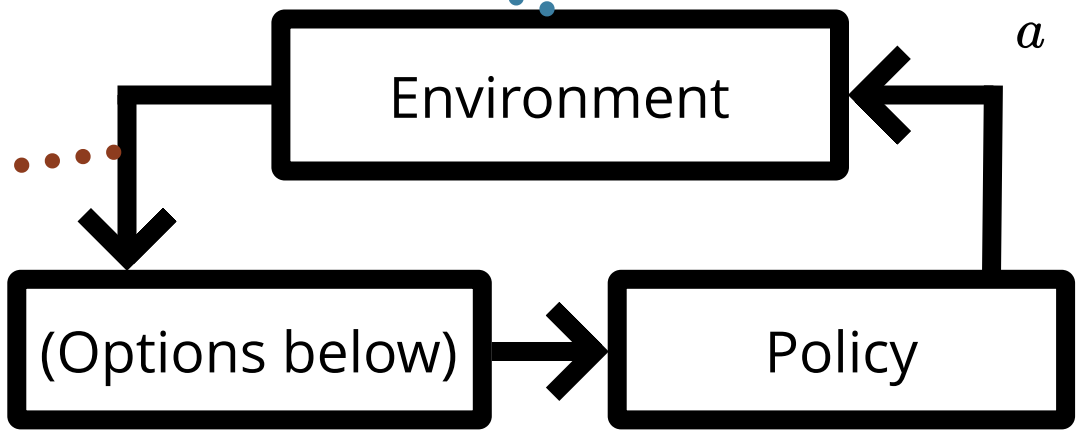
low $N(s,a)/N(s)$ = high bonus

start with $c = 2(\bar{V} - \underline{V})$

How should we adapt MCTS for POMDPs?

True State

$s = TL$



Observation

$o = TL$

Option 1: History

h

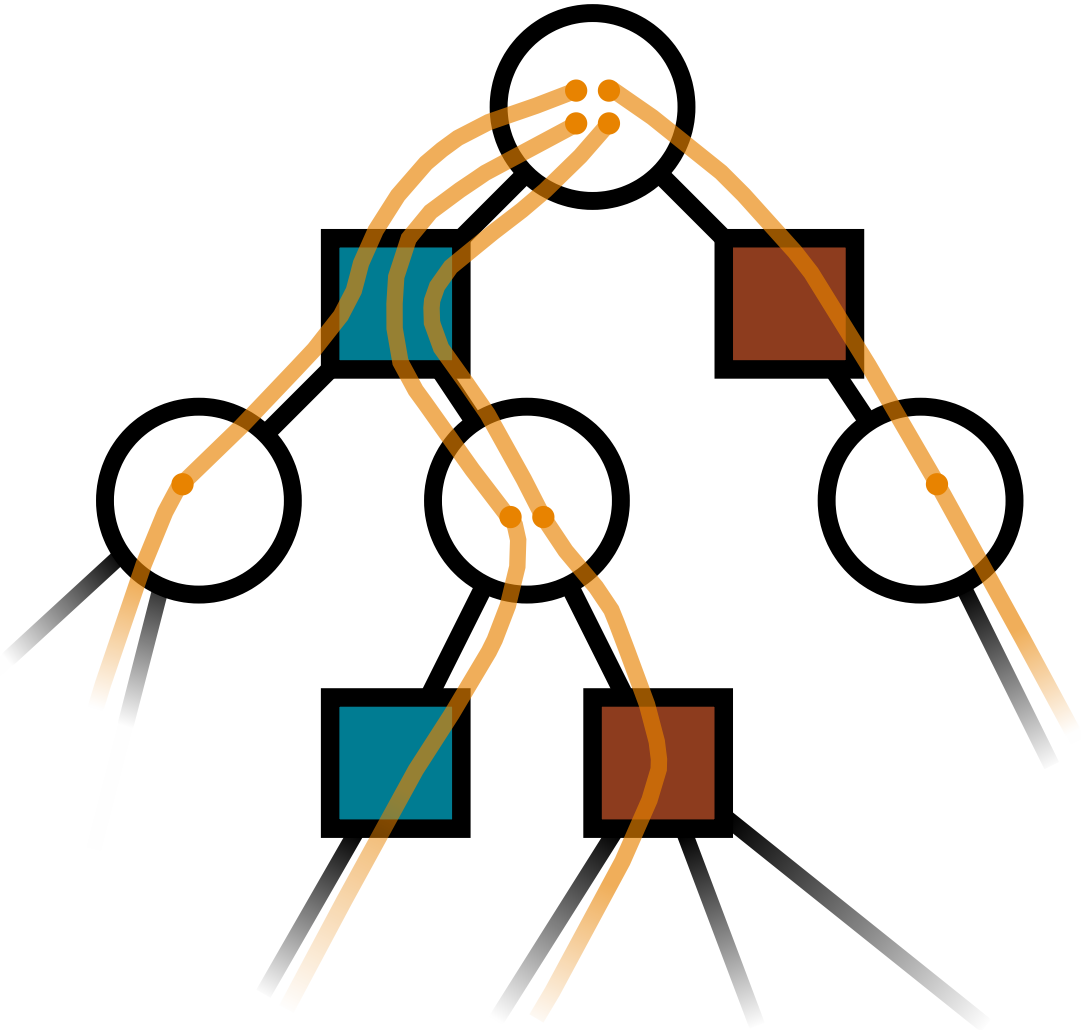
History: $h_t =$

$(b_0, a_0, o_1, a_1, \ldots a_{t-1}, o_t)$

MCTS on Histories

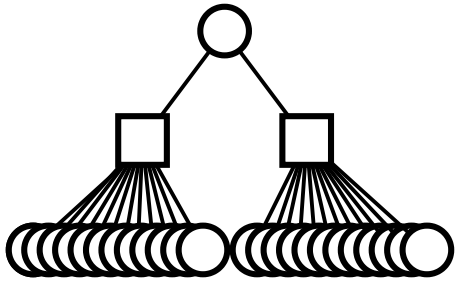
Algorithm 6.12 Monte Carlo tree search

```
function SelectAction(b, d)
    h ← ∅
    loop
        s ~ b
        Simulate(s, h, d)
    return arg max_a Q(h, a)
function Simulate(s, h, d)
    if d = 0
        return 0
    if h ∉ T
        for a ∈ A(s)
            (N(h, a), Q(h, a)) ← (N_0(h, a), Q_0(h, a))
        T = T ∪ {h}
        return Rollout(s, d, π_0)
    a ← arg max_a [ Q(h, a) + c √( log N(h) / N(h, a) ) ]
    (s′, o, r) ~ G(s, a)
    q ← r + γ Simulate(s′, hao, d − 1)
    N(h, a) ← N(h, a) + 1
    Q(h, a) ← Q(h, a) + (q − Q(h, a)) / N(h, a)
    return q
```

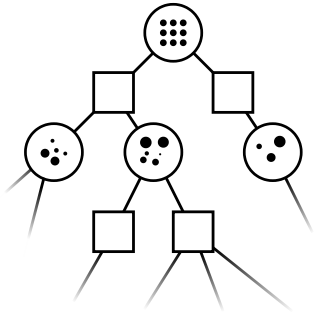


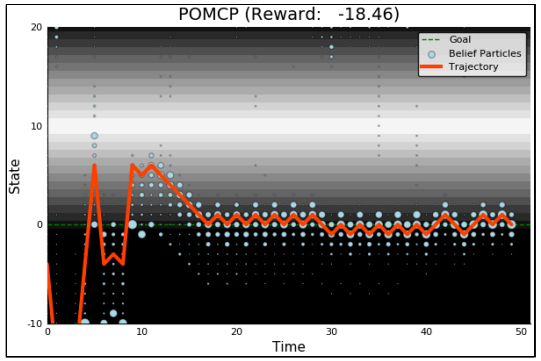
Continuous Observation Spaces

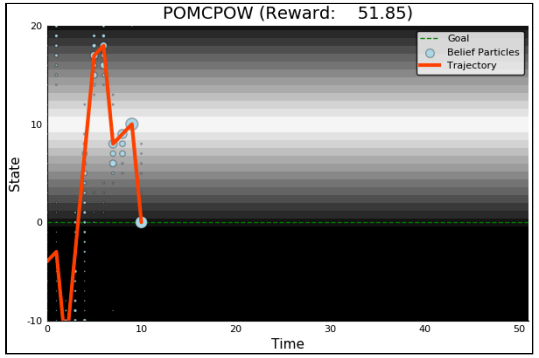
POMCP



POMCPOW





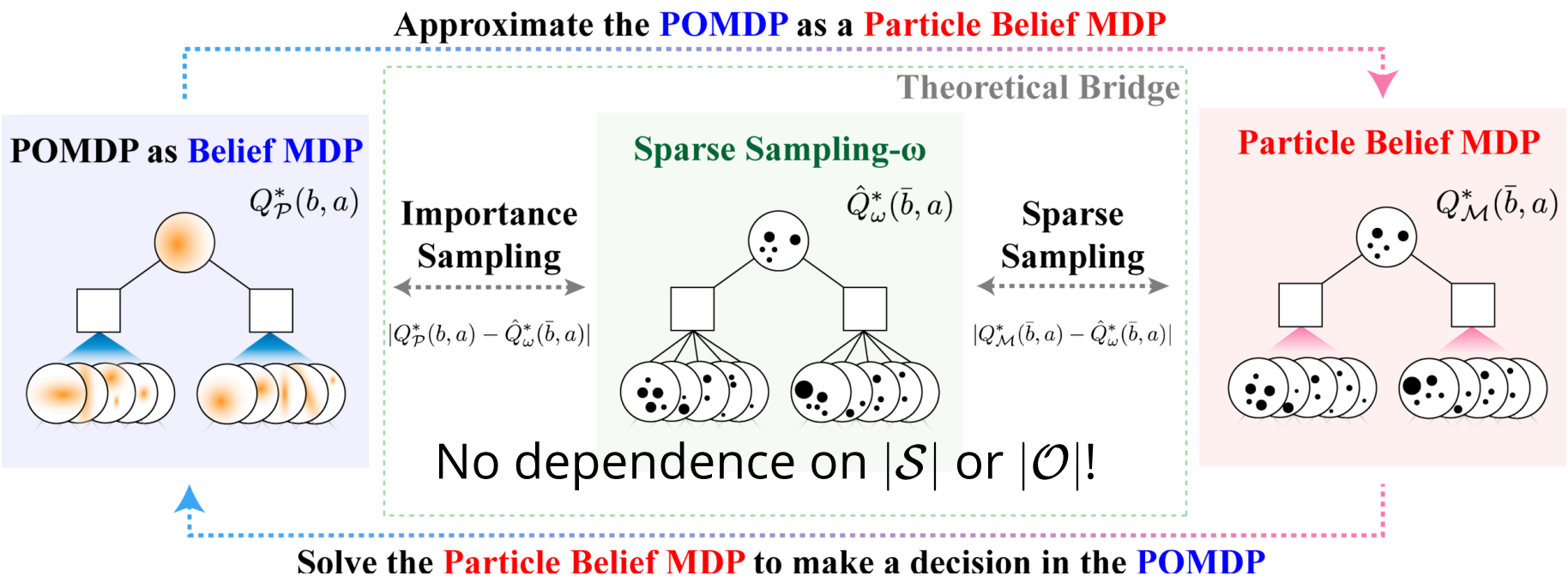


PF Approximation Accuracy

$\mathbf{M_P}$ = Particle belief MDP approximation of POMDP $\mathbf{P}$

For any $\epsilon > 0$ and $\delta > 0$, if C (number of particles) is high enough,

$$|Q^*_{\mathbf{P}}(b,a) - Q^*_{\mathbf{M_P}}(\bar{b},a)| \leq \epsilon \quad \text{w.p. } 1-\delta$$



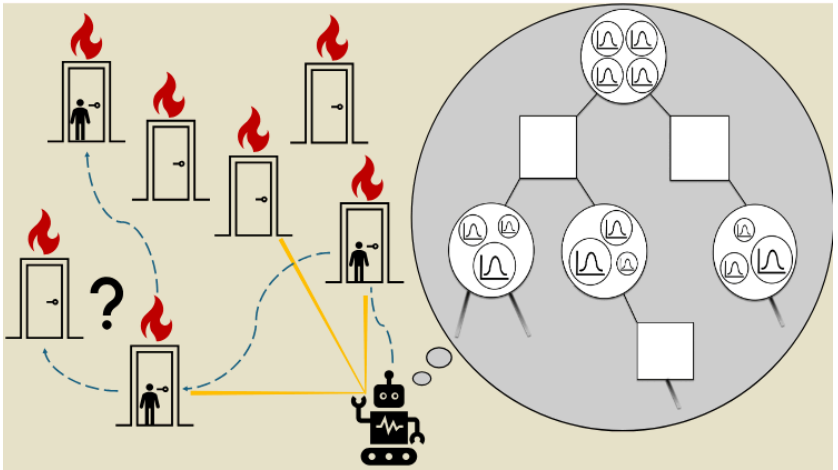
[Lim, Becker, Kochenderfer, Tomlin, & Sunberg, 2023]

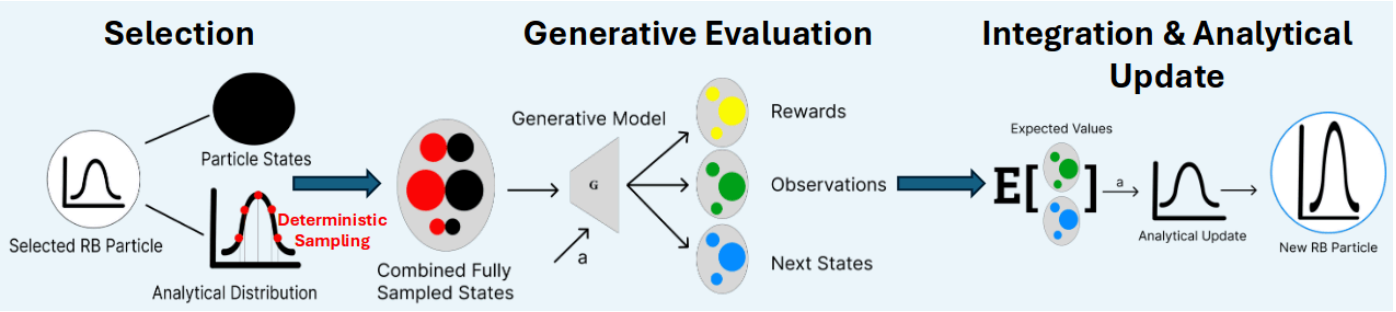
Two Caveats

1. Number of particles is too big for practical safety guarantees.
2. Depends on the Renyi Divergence $d_\infty(\mathcal{P}\|\mathcal{Q})$, which often grows in high-dimensional problems.

Possible Solution to 2:

Rao-Blackwellized POMDP Planning





$$p(s_t \mid h_t)$$

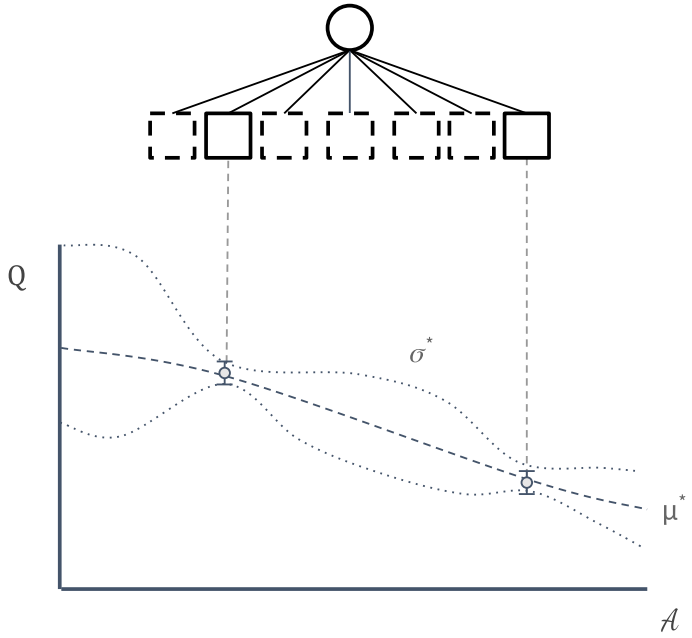
$$= \underbrace{p(s_t^a \mid s_t^p, h_t)}_{\text{Analytical update (e.g. Kalman Filter)}} \underbrace{p(s_t^p \mid h_t)}_{\text{Particle Filter}}$$

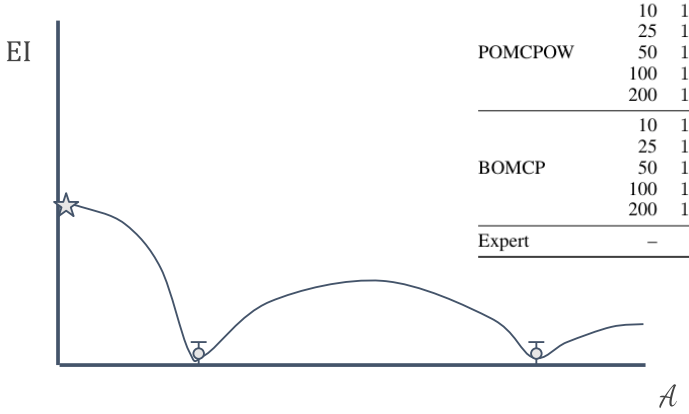
[Lee, Wray, Sunberg, Ahmed, ICRA 2026]

Continuous Action Spaces

BOMCP

Bayesian Optimized Action Branching





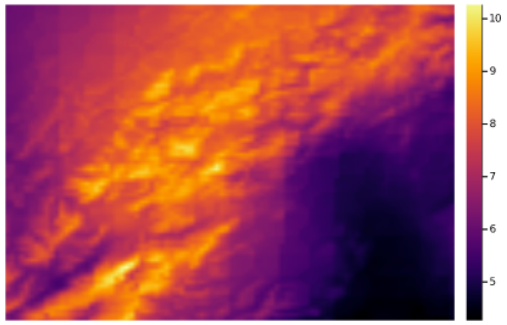
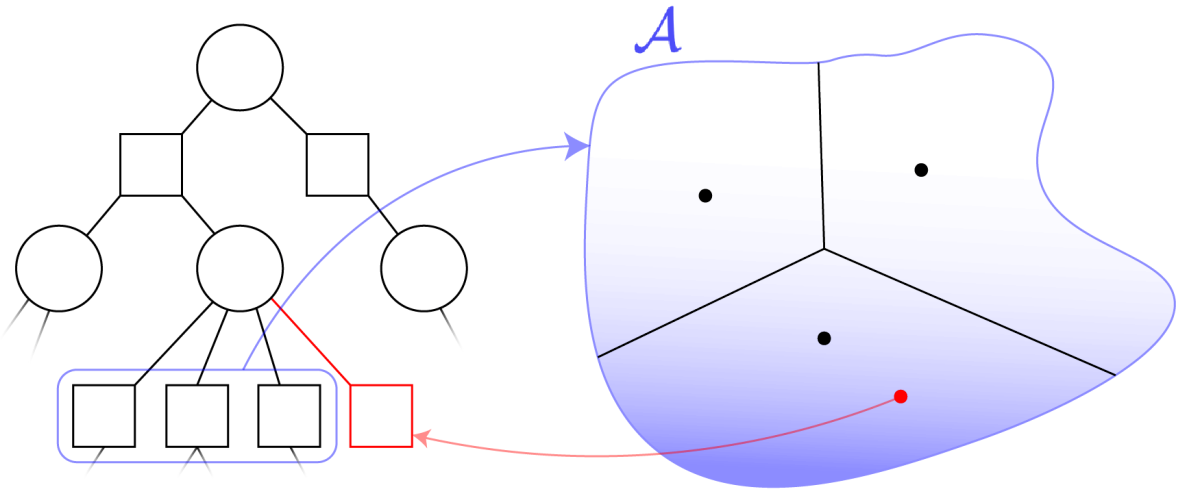
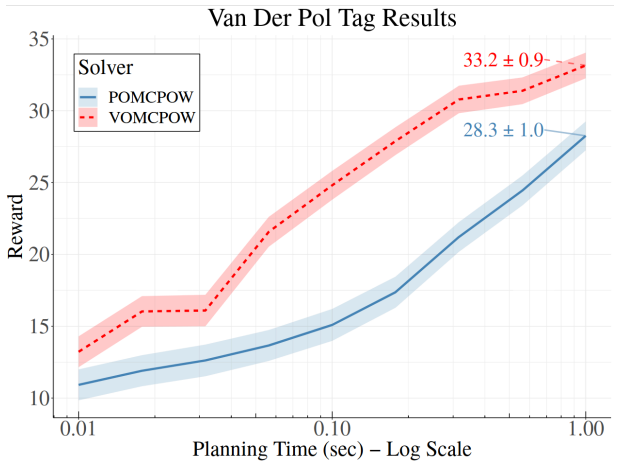


Figure 2: Wind Map. Figure shows wind map for Altamont Pass, CA at 100m altitude. The colors represent the average annual wind speed in m/s.

Algorithm	Queries	Score	Time (seconds)
POMCPOW	10	15708 ± 229	2.25 ± 0.07
	25	16234 ± 217	4.80 ± 0.07
	50	16374 ± 212	6.27 ± 0.08
	100	16018 ± 262	11.98 ± 0.07
	200	15787 ± 233	20.67 ± 0.09
BOMCP	10	18095 ± 183	2.55 ± 0.08
	25	18154 ± 158	5.21 ± 0.07
	50	18015 ± 163	6.71 ± 0.06
	100	18225 ± 119	13.39 ± 0.07
	200	18113 ± 157	25.14 ± 0.08
Expert	–	8130 ± 51	–

Voronoi Progressive Widening



Online Tree Search Planner Voronoi Progressive Widening